for some constants $U_{\min}$ and $U_{\max}$. Hence ϕ is strictly positive over the entire space, and we can compute that

$$\begin{aligned}\mathcal{L}_{\mathrm{NHL}}\phi &= \boldsymbol{p}\cdot\nabla U(\boldsymbol{q}) - \nabla U(\boldsymbol{q})\cdot\boldsymbol{p} - \xi\|\boldsymbol{p}\|^2 + \left[\|\boldsymbol{p}\|^2 - \mathrm{N_d k_B T}\right](\xi - a)\\ &\quad - \gamma\mu\xi(\xi - a) + \mathrm{k_B T}\gamma,\\ &\leq -a\|\boldsymbol{p}\|^2 - \gamma\mu\xi^2 + (a\gamma\mu - \mathrm{N_d k_B T})\,\xi + a\mathrm{N_d k_B T} + \mathrm{k_B T}\gamma.\end{aligned}$$

Choosing a so that $a\gamma\mu - \mathrm{N_d k_B T} = 0$, we obtain

$$\mathcal{L}_{\mathrm{NHL}}\phi = -a\|\boldsymbol{p}\|^2 - \gamma\mu\xi^2 - \gamma\mu a^2 + \gamma\mu a^2 + a\mathrm{N_d k_B T} + \mathrm{k_B T}\gamma.$$

Using $-\xi^2 - a^2 \leq -(\xi - a)^2/2$ gives

$$\mathcal{L}_{\mathrm{NHL}}\phi \leq -2a\frac{1}{2}\|\boldsymbol{p}\|^2 - \gamma\mu\frac{1}{2}(\xi - a)^2 + \gamma\mu a^2 + a\mathrm{N_d k_B T} + \mathrm{k_B T}\gamma.$$

Now choosing

$$\alpha = \min\left(2a, \gamma\mu\right),$$

we have

$$\begin{aligned}\mathcal{L}_{\mathrm{NHL}}\phi &\leq -\alpha\phi + \alpha U(\boldsymbol{q}) + \gamma\mu a^2 + a\mathrm{N_d k_B T} + \mathrm{k_B T}\gamma,\\ &\leq -\alpha\phi + \delta,\end{aligned}$$

for constants α and δ as required, where

$$\delta = \alpha U_{\max} + \gamma\mu a^2 + a\mathrm{N_d k_B T} + \mathrm{k_B T}\gamma.$$

This gives the necessary Lyapunov condition for the Nosé-Hoover-Langevin equations, and hence the dynamics are geometrically ergodic. Solution trajectories will sample the extended canonical distribution, and it remains to compute solutions using an appropriate discretization scheme.

8.4.2 Numerical Integration of the Nosé-Hoover-Langevin Equations

Many older schemes for Nosé-Hoover and Langevin dynamics make use of computational shortcuts to (for example) reduce the number of computed random values per iteration, reduce the data storage needed for an algorithm or replace functions that are costly to evaluate (e.g. exponential or trigonometric functions)